

# Worksheet

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1. (The idea for this exercise is from Liza Bolton.) The following code makes a nice graph. Modify it to create the ugliest graph that you can.

```
library(ggplot2)

ggplot(iris, aes(x = Sepal.Length, y = Sepal.Width, colour = Species)) +
  geom_point(size = 2.5, alpha = 0.8) +
  labs(
    title = "Iris sepal dimensions",
    x = "Sepal length (cm)",
    y = "Sepal width (cm)",
    colour = "Species"
  ) +
  theme_minimal()
```

2. In groups of two, pick one of this week's readings and make a two-to-five slide summary of its most important points using Quarto. Make a PR that adds your slides to the class GitHub repo.
  - Timbers et al., *Reproducible and Trustworthy Workflows for Data Science*, the version control chapters.
  - Polars User Guide, “Getting started” and “Concepts”, together with the Parquet page of the Apache Arrow Python docs.
  - Healy, *Data Visualization*, Chapter 1.
  - Alexander, *Telling Stories with Data*, Chapter 5.
3. Make a Quarto website and put it online. Create the project (`quarto create project website`), add a page with one graph of the shelter data from Week 1, set `output-dir: docs` in `_quarto.yml`, render, and push to a new GitHub repository. Under Settings, then Pages, deploy from `main` and the `/docs` folder. Write down the URL and open it on someone else's laptop.

4. Make three graphs from three different Open Data Toronto datasets using ggplot2, and one table using tinytable, and add them as a page on your website.